MINI PROJECT — *Generative AI* by **IBM**

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CSE AI-A

CYBERSECURITY

Comprehensive University-Level Study Material

Beginner → Advanced → Professional

Theory • Practical Skills • Security Analysis • Professional Practice

Preface

This study material is designed as a structured foundation for students entering cybersecurity. It treats cybersecurity as an academic and professional discipline rather than as a collection of hacking tools or commands.

The material follows a systems-first approach: understand computing, operating systems, networks, programming and applications before studying attacks and defenses. Each topic is framed around what the technology is, why it exists, how it works, what can go wrong, how attacks can be detected, and how systems can be secured.

All practical security testing described in this material should be performed only on systems owned by the learner or on systems for which explicit authorization has been granted.

How to Use This Material

1. Learn the concepts and terminology before using security tools.
2. Draw the architecture or data flow whenever a topic feels abstract.
3. For every vulnerability, learn both the root cause and the mitigation.
4. Practice only in controlled, authorized laboratory environments.
5. Use the review questions to test understanding rather than memorization.
6. Keep a laboratory notebook containing observations, evidence, and lessons learned.

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PART I — COMPUTING FOUNDATIONS

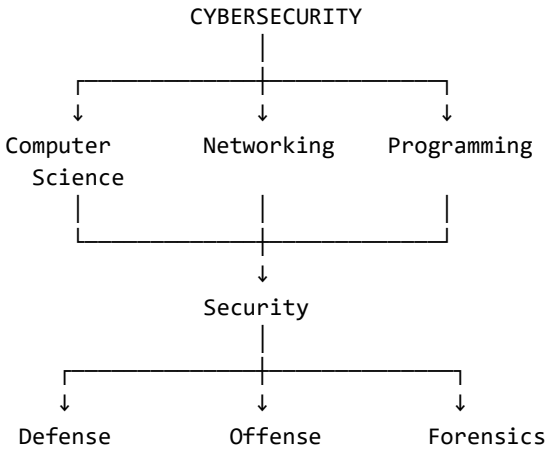
Cybersecurity begins with understanding the systems being protected. The following chapters establish the computing, operating-system, programming and networking foundations required for later security study.

Chapter 1 — Introduction to Cybersecurity

1.1 What is Cybersecurity?

Cybersecurity is the discipline concerned with protecting information systems, networks, applications, devices and data against unauthorized access, misuse, disruption, modification or destruction.

A cybersecurity professional therefore needs to understand both how systems work and how systems can fail.



1.2 What Are We Protecting?

An asset is something valuable that requires protection. Assets can include personal information, credentials, intellectual property, databases, servers, applications, cloud resources, network infrastructure and organizational reputation.

1.3 The CIA Triad

Principle	Meaning	Example
Confidentiality	Only authorized parties should access information.	A student's academic record is not publicly visible.
Integrity	Information should remain accurate and trustworthy.	A student cannot improperly change an examination mark.
Availability	Authorized users should be able to access resources when needed.	An examination portal remains accessible during registration.

Memory aid: Confidentiality = keep it secret; Integrity = keep it correct; Availability = keep it accessible.

1.4 Security Beyond CIA

- Authenticity — establishing that an entity or piece of information is genuine.

- Accountability — being able to associate actions with responsible identities or systems.
- Non-repudiation — supporting evidence that an action or transaction occurred and can be attributed appropriately.
- Privacy — handling personal information appropriately and according to applicable requirements.

1.5 Threat, Vulnerability, Exploit and Risk

Term	Meaning
Threat	Something capable of causing harm to an asset.
Vulnerability	A weakness that could contribute to security harm.
Exploit	A technique, mechanism or code that takes advantage of a vulnerability.
Risk	The potential for loss or harm arising from threats and weaknesses.

```
Weakness
   ↓
Vulnerability
   ↓
Exploitation
   ↓
Unauthorized action
   ↓
Impact
```

1.6 Attack Surface and Attack Vector

The attack surface is the collection of points at which an unauthorized party might interact with a system. An attack vector is a pathway through which an attack may occur, such as phishing, stolen credentials, vulnerable applications or insecure configuration.

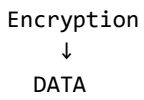
1.7 Security Controls

- Preventive controls — designed to reduce the chance of an incident, such as MFA and access controls.
- Detective controls — designed to identify suspicious activity, such as logging, IDS and monitoring.
- Corrective controls — designed to restore or correct systems, such as recovery procedures and remediation.

1.8 Defense in Depth

Defense in depth means using multiple complementary security layers rather than relying on a single control.

```
ATTACK
   ↓
Firewall
   ↓
Segmentation
   ↓
MFA
   ↓
Access Control
   ↓
```



1.9 Security Mindset

7. What are we protecting? → Assets
8. Who should have access? → Authorization
9. How do we know who they are? → Authentication
10. What could go wrong? → Threat modeling
11. What weaknesses exist? → Vulnerability analysis
12. How would we detect an attack? → Monitoring
13. How would we stop or reduce it? → Security controls
14. What happens if prevention fails? → Incident response

Chapter 1 Case Study

Imagine a university with a student database containing names, student IDs, marks, attendance and other personal information. The database is an asset. Potential threats include external attackers, malicious insiders, phishing and malware. Potential weaknesses might include weak passwords, lack of MFA, excessive permissions and insufficient logging.

A security professional could recommend MFA, strong access controls, network segmentation, secure password storage, monitoring, backups and incident-response procedures.

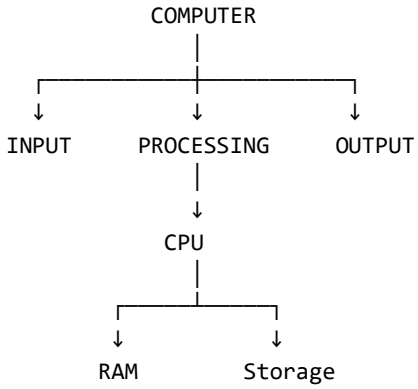
The central reasoning pattern is: Asset → Threat → Vulnerability → Risk → Control.

Chapter 1 Review Questions

15. What are the three components of the CIA Triad?
16. What is the difference between authentication and authorization?
17. Give an example of an asset, a threat, a vulnerability and a security control.
18. Why is defense in depth important?
19. Explain the relationship Asset → Threat → Vulnerability → Risk → Control.

Chapter 2 — Computer Architecture

2.1 Basic Computer Model



2.2 CPU

The Central Processing Unit executes instructions. Important components include the Arithmetic Logic Unit (ALU), the control unit and registers.

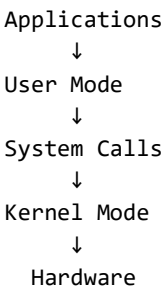
2.3 RAM and Storage

RAM provides temporary working memory for active programs and is generally volatile. Persistent storage such as SSDs and HDDs retains information when power is removed.

2.4 Security Relevance

Malicious software can exist as a file on storage, be loaded into memory, execute as a process and communicate over a network. Consequently, security investigations often need to consider storage, memory, processes and network activity together.

2.5 Privilege



The kernel generally has greater authority than ordinary applications. This leads to the principle of least privilege: a process or user should receive only the permissions necessary to perform its legitimate function.

Chapter 2 Review

- CPU
- RAM
- Storage
- Registers
- Kernel
- User mode
- Privileged execution
- Least privilege

Chapter 3 — Operating Systems

An operating system manages computer hardware and provides services to applications. Examples include Windows, Linux, macOS, Android and iOS.

3.1 Major Responsibilities

- Process management
- Memory management
- File management

- Device management
- User and permission management
- Networking

3.2 Processes

```
Program
  ↓
Execution
  ↓
Process
```

A running process uses resources such as memory, CPU time, file handles, network connections and permissions. Security analysts may investigate unexpected processes, but suspicious behavior must always be interpreted in context.

3.3 Users, Groups and Permissions

Operating systems use identities and groups to determine access to resources. In Unix-like systems, a simplified file permission model uses owner, group and others, with read (r), write (w) and execute (x) permissions.

```
rwx r-x r--
Owner:  read/write/execute
Group:  read/execute
Others: read
```

Permissions are a practical implementation of least privilege.

Chapter 4 — Linux Fundamentals

Linux is widely used in servers, cloud infrastructure, security appliances, development environments, research and security laboratories.

4.1 Filesystem

```
/
├── home
│   ├── alice
│   └── bob
├── etc
├── var
├── tmp
├── usr
└── opt
```

4.2 Essential Commands

Command	Purpose
pwd	Show current directory
ls	List files
cd	Change directory

mkdir	Create directory
touch	Create file
cp	Copy
mv	Move or rename
rm	Remove
cat	Display file contents
less	View a file
grep	Search text
find	Find files
chmod	Modify permissions
chown	Change ownership
ps	View processes
ip	View network information
ss	View network sockets

4.3 Linux Security Concepts

- Users and groups
- File permissions
- sudo
- Processes
- Services
- Logs
- SSH
- Firewall configuration
- Package management

Chapter 5 — Programming for Cybersecurity

Programming improves the ability to automate tasks, analyze logs, process data, understand applications, read tools, investigate malware and build defensive systems.

5.1 Python Learning Sequence

```
Variables
   ↓
Data Types
   ↓
Conditions
   ↓
Loops
   ↓
Functions
   ↓
Collections
```

```
   ↓
Files
   ↓
Exceptions
   ↓
Modules
   ↓
APIs
   ↓
Automation
```

5.2 Security-Oriented Programming

A useful beginner project is a log-analysis program that reads authorized log data, identifies authentication failures, counts events and reports patterns for further investigation.

5.3 Skills to Develop

- Beginner: variables, conditions, loops, functions.
- Intermediate: file processing, exceptions, modules, JSON, APIs, regular expressions.
- Advanced: networking, concurrency, system interaction, databases and automation.

Chapter 6 — Computer Networks

Networking is one of the most important foundations of cybersecurity. Many security events are fundamentally communication events.

```
Laptop
   |
   ↓
Router
   |
   ↓
Internet
   |
   ↓
Web Server
```

6.1 IP Addresses

An IP address identifies a network interface for communication at the IP layer. Students should learn IPv4, IPv6, public and private addressing, loopback, subnet masks and CIDR.

6.2 Ports

A port identifies a service endpoint. A useful analogy is: IP address = building; port = particular door or service.

Port	Common Service
22	SSH
53	DNS
80	HTTP

6.3 TCP and UDP

TCP provides reliable, ordered communication with mechanisms such as connection establishment, acknowledgments, retransmission and flow control. UDP is a simpler connectionless transport mechanism with lower overhead and different reliability characteristics.

Chapter 7 — The OSI Model

```
7 — Application
6 — Presentation
5 — Session
4 — Transport
3 — Network
2 — Data Link
1 — Physical
```

Mnemonic: All People Seem To Need Data Processing.

The OSI model is a conceptual framework for understanding network communication. Security analysts can use the layers as a systematic way to reason about where an issue occurs.

Chapter 7 Review

20. Name the seven OSI layers.
21. Which layer is associated with TCP and UDP?
22. Why is a layered model useful when investigating network behavior?

Chapter 8 — DNS

The Domain Name System maps human-readable names to network information and supports many kinds of records used by Internet services.

```
Browser
   ↓
"example.com"
   ↓
DNS
   ↓
IP address
   ↓
Server
```

DNS is important to security because it is fundamental to web access, enterprise infrastructure, email, threat detection and incident investigation.

Chapter 9 — HTTP & HTTPS

```
Browser
   |
```

```
    | HTTP request
    ↓
Web Server
    |
    | HTTP response
    ↓
Browser
```

9.1 HTTP Methods

Method	Typical Purpose
GET	Retrieve a resource
POST	Submit data or request creation
PUT	Replace or update a resource
PATCH	Partially modify a resource
DELETE	Request deletion

9.2 HTTP Status Codes

Code	Meaning
200	Successful request
201	Resource created
301	Redirect
400	Bad request
401	Authentication required/failed
403	Forbidden
404	Not found
500	Server-side error

9.3 HTTPS

HTTPS is HTTP protected by TLS. TLS is designed to provide properties including confidentiality, integrity and authentication of the server under the applicable trust model.

Chapter 10 — Authentication & Authorization

```
Authentication
      ↓
Identity

Authorization
      ↓
Permission
```

Authentication answers 'Who are you?' Authorization answers 'What are you allowed to do?'

10.1 Authentication Factors

- Something you know — password or PIN.
- Something you have — security key or authenticator device.
- Something you are — biometric characteristic.

Using multiple factor categories is called multi-factor authentication (MFA).

Chapter 11 — Cryptography

Cryptography provides mechanisms for protecting information and communications.

11.1 Encryption

```
Plaintext
   ↓
Encryption + Key
   ↓
Ciphertext
   ↓
Decryption
   ↓
Plaintext
```

11.2 Symmetric Cryptography

Symmetric encryption uses a secret key for encryption and decryption. AES is a widely used modern example.

11.3 Asymmetric Cryptography

Asymmetric cryptography uses a key pair consisting of a public key and a private key. Applications include digital signatures and key establishment.

11.4 Hashing

A cryptographic hash function maps input data to a fixed-length digest. Hashing is not encryption. Examples include SHA-256 and SHA-3.

11.5 Password Storage

Passwords should not be stored as plaintext. Secure systems use dedicated password-hashing schemes with appropriate salts and work factors, such as Argon2, bcrypt or scrypt.

Chapter 12 — Web Application Security

```
USER
  ↓
Browser
  ↓
Web Server
  ↓
Application
```

```
        ↓
    Database
```

12.1 Input Validation

Applications should treat external input as untrusted until it has been appropriately validated and safely handled. Inputs may arrive through forms, URLs, headers, cookies, APIs and file uploads.

12.2 SQL Injection

SQL injection occurs when untrusted input is improperly incorporated into database queries so that data is confused with SQL syntax. Primary defenses include parameterized queries, prepared statements, suitable validation and least-privilege database accounts.

12.3 Cross-Site Scripting

XSS occurs when untrusted content is improperly handled so that it can be interpreted as executable content in another user's browser. Categories include reflected, stored and DOM-based XSS.

Defenses include context-aware output encoding, safe templating, appropriate input handling and Content Security Policy.

12.4 Broken Access Control

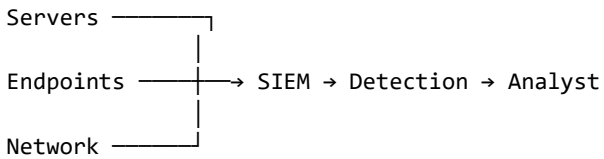
Broken access control occurs when an application allows users to perform actions or access resources beyond their authorization. Authorization must be enforced server-side for protected operations.

Chapter 13 — Security Operations

Security operations continuously monitor systems, investigate alerts and improve defensive controls.

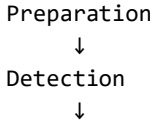
13.1 Logs

- Authentication events
- Process activity
- Network connections
- Application errors
- Administrative actions



A SIEM can collect and correlate security-related events to support monitoring and investigation.

Chapter 14 — Incident Response



```
Analysis
     ↓
Containment
     ↓
Eradication
     ↓
Recovery
     ↓
Lessons Learned
```

Incident response is a structured process for handling suspected or confirmed security incidents. The investigation should establish what happened, when it happened, which accounts and systems were involved, what evidence exists, what needs containment and how recovery should occur.

Chapter 15 — Digital Forensics

Digital forensics is the systematic examination of digital evidence. Evidence can come from computers, phones, servers, cloud environments, network traffic, memory and storage.

15.1 Forensic Principles

- Preservation
- Documentation
- Analysis
- Validation
- Reporting

Evidence should be handled in a way that maintains its integrity and supports reliable conclusions.

Chapter 16 — Malware

Category	General Description
Trojan	Malicious software disguised as something legitimate.
Worm	Malware capable of spreading between systems through network or other mechanisms.
Ransomware	Malware designed to deny access to data or systems, commonly through encryption, while demanding payment.
Spyware	Software designed to collect information without appropriate authorization.
Rootkit	Software designed to conceal activity or maintain privileged access.

Chapter 17 — Cloud Security

17.1 Service Models

- IaaS — Infrastructure as a Service.
- PaaS — Platform as a Service.
- SaaS — Software as a Service.

17.2 Shared Responsibility

Cloud security responsibilities are divided between the provider and customer, with the exact division depending on the service model.

17.3 Major Security Areas

- Identity and access management
- Network controls
- Storage security
- Encryption
- Secrets management
- Logging
- Monitoring
- Configuration management
- Backup
- Incident response

Chapter 18 — Ethical Hacking

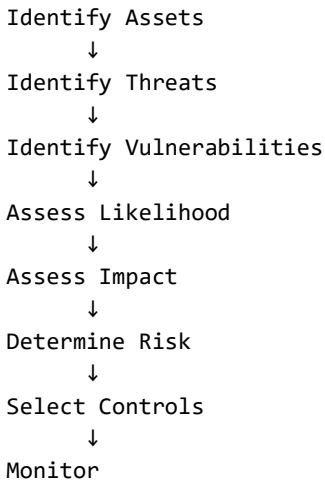
Ethical hacking means conducting authorized security testing. Authorization and scope are essential professional requirements.

```
Planning
   ↓
Reconnaissance
   ↓
Enumeration
   ↓
Vulnerability Analysis
   ↓
Controlled Validation
   ↓
Impact Assessment
   ↓
Reporting
   ↓
Remediation
   ↓
Retesting
```

A professional assessment is more than running tools. It requires a defined scope, methodology, evidence, risk analysis, remediation guidance and retesting.

Chapter 19 — Security Governance & Risk

Organizations must determine what needs protection, how much protection is appropriate, how much risk is acceptable, what security controls are justified and what legal or regulatory requirements apply.



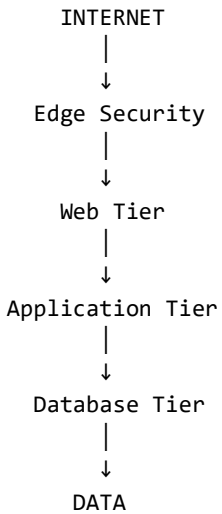
Chapter 20 — Zero Trust

Zero Trust rejects the assumption that being inside a network automatically makes a user or device trustworthy.

Core principles include verifying identity, applying least privilege, considering device and contextual information, segmenting resources, continuously monitoring activity and assuming that compromise is possible.

Chapter 21 — Security Architecture

Security architecture asks how an entire system should be designed so that security is built into its structure rather than added after the fact.



UNIVERSITY-LEVEL PRACTICAL PROGRAM

Practical work should be performed only in owned, deliberately vulnerable or explicitly authorized environments. The purpose of the laboratory is to develop reasoning, investigation and defensive skills—not to target real systems.

Lab 1 — Linux Fundamentals

Objectives

- Navigate Linux
- Create and manage files
- Manage permissions
- Identify processes
- Examine logs

Lab 2 — Network Analysis

Objectives

- Identify communicating hosts in authorized traffic
- Identify protocols and ports
- Construct a basic event timeline
- Explain observed communication

Lab 3 — Secure Programming

Objectives

- Identify a deliberately insecure design
- Explain the root cause
- Implement a correction
- Verify the correction

Lab 4 — Web Security

Objectives

- Study authentication and authorization
- Analyze input handling
- Examine session behavior
- Identify and remediate common security weaknesses

Lab 5 — SOC Investigation

Objectives

- Analyze simulated authentication logs
- Build a timeline
- Develop a hypothesis

- Assess severity
- Recommend actions

Lab 6 — Digital Forensics

Objectives

- Inventory evidence
- Construct a timeline
- Document findings
- Assess confidence
- Produce a forensic report

Lab 7 — Incident Response

Objectives

- Work in incident-response roles
- Investigate a simulated compromise
- Recommend containment
- Communicate findings

ASSESSMENT FRAMEWORK

Component	Weight
Theory examinations	25%
Laboratory work	20%
Assignments	10%
Security projects	15%
Research paper	10%
Practical examination	10%
Capstone	10%

Assessment should test conceptual understanding, practical competence, analytical reasoning, professional communication and the ability to design or evaluate security controls.

HOW TO STUDY EACH CHAPTER

23. KNOW — Can you define the terminology accurately?
24. UNDERSTAND — Can you explain the concept without notes?
25. APPLY — Can you use the concept in an authorized laboratory?
26. ANALYZE — Can you determine why a security problem occurred?
27. DESIGN — Can you design an appropriate solution?
28. EVALUATE — Can you compare solutions and explain trade-offs?

The Ten-Question Security Framework

29. What is it?
30. Why does it exist?
31. How does it work?
32. What are its components?
33. What can go wrong?
34. What vulnerabilities can affect it?
35. How could an attacker abuse those weaknesses?
36. How can we detect the abuse?
37. How can we prevent or mitigate it?
38. How would we investigate an incident involving it?

FIRST-SEMESTER STUDY PLAN

Week	Topic
1	Introduction to Computing
2	Computer Architecture
3	Operating Systems
4	Linux Fundamentals
5	Linux Administration
6	Programming Fundamentals
7	Python for Security
8	Data Structures
9	Networking Fundamentals
10	OSI & TCP/IP
11	IP Addressing
12	TCP/UDP & Ports
13	DNS, DHCP & ARP
14	HTTP/HTTPS
15	Network Security
16	Practical Examination & Revision

SECOND SEMESTER — CORE SECURITY

```
Cybersecurity Foundations
        ↓
CIA Triad
        ↓
Threats & Vulnerabilities
        ↓
Risk
```

```
        ↓
Threat Modeling
        ↓
Authentication
        ↓
Authorization
        ↓
Access Control
        ↓
Cryptography
        ↓
PKI
        ↓
Web Security
        ↓
Secure Programming
```

THIRD SEMESTER — SECURITY OPERATIONS

```
Ethical Hacking
      +
Network Security
      +
Security Operations
      +
SIEM
      +
Incident Response
      +
Digital Forensics
```

FOURTH SEMESTER — ADVANCED SECURITY

```
Cloud Security
      +
Malware Analysis
      +
Reverse Engineering
      +
Threat Intelligence
      +
Vulnerability Research
      +
Security Architecture
```

SPECIALIZATION TRACKS

Track	Core Progression
Offensive Security	Network Security → Web Security → Pentesting → Vulnerability Research → Red Teaming

Defensive Security	Networking → Logging → SIEM → Detection Engineering → Incident Response → Threat Hunting
Digital Forensics & IR	OS → Filesystems → Forensics → Memory Analysis → Network Forensics → Incident Response
Cloud Security	Networking → Cloud Fundamentals → IAM → Cloud Architecture → Container Security → Cloud Response
Application Security	Programming → Databases → Web Development → Secure Coding → AppSec → DevSecOps
Malware & Reverse Engineering	Programming → OS → Assembly → Reverse Engineering → Malware Analysis → Threat Research

THE CYBERSECURITY MINDSET

The objective of cybersecurity education is not to memorize a list of tools. The objective is to develop the ability to reason about systems and security.

```
COMPUTER SCIENCE
       ↓
SYSTEMS
       ↓
NETWORKS
       ↓
APPLICATIONS
       ↓
SECURITY PRINCIPLES
       ↓
THREATS
       ↓
VULNERABILITIES
       ↓
ATTACKS
       ↓
DETECTION
       ↓
DEFENSE
       ↓
INVESTIGATION
       ↓
SECURE DESIGN
       ↓
RESEARCH
```

A strong cybersecurity student can look at an unfamiliar system and systematically ask: What are the assets? What are the trust boundaries? What are the attack surfaces? What could go wrong? Why could it happen? How could we detect it? How could we prevent it? How would we respond? How do we verify the fix?

Professional Standard

Knowing the definition of SQL injection is different from understanding why it occurs, recognizing vulnerable design, safely demonstrating the issue in an authorized environment, explaining the impact, implementing the mitigation and verifying that the vulnerability has been addressed. The same standard applies to networking, authentication, cryptography, incident response and every other major cybersecurity discipline.

Closing Principle

Cybersecurity is a continuous discipline. Technology changes, threats evolve and organizations change. A strong professional therefore combines technical foundations, analytical thinking, ethical judgment, communication skills and a habit of continuous learning.

CORE GLOSSARY

Term	Definition

Asset	Something valuable that requires protection.
Authentication	Process of establishing an identity.
Authorization	Process of determining what an authenticated identity is permitted to do.
Availability	Property that authorized users can access resources when required.
Confidentiality	Property that information is accessible only to authorized parties.
Exploit	Technique or mechanism that takes advantage of a vulnerability.
Integrity	Property that information remains accurate and trustworthy.
Risk	Potential for loss or harm arising from threats and weaknesses.
Threat	Something capable of causing harm.
Vulnerability	A weakness that may contribute to security harm.
Zero Trust	Security approach that avoids implicit trust and emphasizes continuous verification.

END OF STUDY MATERIAL

Cybersecurity • Foundations to Professional Practice